

3.2 Sub-module 1: Getting Familiar with Generative AI (GenAI)

This sub-module teaches students to produce content with generative AI tools to create design artefacts [GenAI-Obj]. Students develop procedural knowledge through practical guided activities (e.g. Midjourney step-by-step tool tutorials) and they are asked to apply the acquired procedure to a familiar task (e.g. creating an original shape for a lamp inspired by nature). Students are first provided with a basic literacy of AI including technical vocabulary (GenAI-Act1) and primary functionalities (GenAI-Act2). This activity is meant to promote factual and conceptual knowledge by teaching the basic elements of the discipline and their interrelationship to explain the functionality of AI tools. Once basic literacy is introduced, specific case studies are provided to contextualise the application of AI within the design practice (GenAI-Act3).

At this point, the teaching moves from a theoretical to a procedural level. In this direction, students are first introduced to a guided tutorial (ML-Act4.3) on how to use an ML tool (e.g. runway) starting from setting a design goal (ML-Act4.1) and creating their dataset (ML-Act4.2). Then, to familiarise themselves with the procedure, students individually apply what they learned by training an ML model aligned with the design goal (ML-Act5). Lastly, moving the focus to meta-cognitive knowledge, students are asked to analyse and document the process by breaking down the steps and selecting the relevant results (ML-Act6). This activity aims to create awareness and knowledge of cognition. To foster participation and sharing of results, students are asked to present their work and question others (ML-Act7).

Educational Objective {GenAI-Obj} :

Students will learn to generate contents with GenAI tools to create design artifacts.

Sub-module 1: Getting familiar w/ GenAI	The cognitive process dimension					
	(1) Remember	(2) Understand	(3) Apply	(4) Analyze	(5) Evaluate	(6) Create
	(a) Factual Knowledge	{GenAI-Act.1}				
	(b) Conceptual Knowledge	{GenAI-Act.2} {GenAI-Act.3}				
	(c) Procedural Knowledge	{GenAI-Act.4} {GenAI-Act.4.1, 4.2, 4.3}	{GenAI-Obj} {GenAI-Act.5}			
(d) Meta Knowledge				{GenAI-Act.6}	{GenAI-Act.7}	

- {Act.1} Activities intended to provide students with vocabulary knowledge of GenAI;
- {Act.2} Activities intended to provide students with a basic functionality of GenAI;
- {Act.3} Activities intended to provide students with practical examples of GenAI applied to design;
- {Act.4} Activities intended to explain the procedure of generating contents with GenAI;
- {Act.4.1} Activities intended to set or aligning with a design goal;
- {Act.4.2} Activities intended to introduce students in familiarising, collecting or creating prompts (data);
- {Act.4.3} Activities intended to provide a step-by-step tool tutorial, such as Midjourney, Dall-E;
- {Act.5} Activities intended to allow students to individually generate contents with GenAI for a design goal;
- {Act.6} Activities intended to provide students with a structured template for process and results documentation;
- {Act.7} Activities intended to make students explaining and questioning the process;